

The empirical recurrence for OEIS A278017: recovering a conjecture that is not written down

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Abstract

OEIS A278017 records “Empirical recurrence of order 63”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 114$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 63, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 63 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A278017 is “Number of $4 \times n$ 0..2 arrays with every element plus 1 mod 3 equal to some element at offset (-1,-1) (-1,0) (-1,1) (0,-1) (0,1) or (1,0), with upper left element zero.”. It has offset 1 and begins

0, 198, 7204, 315060, 13597418, 586416930, 25289030636, 1090585845484, ...

The entry states:

Empirical recurrence of order 63 (see link above)

As of the “Last modified” line on the live entry (Nov 08 2016 12:33:22, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 6561$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 114$ of the 6561, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 114$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 228$ exact terms of the model returns order 63 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{63} c_i a(n-i),$$

c_1	54	c_{17}	−162273446402004	c_{33}	−481170644079430767560	c_{49}	−9547344509485845
c_2	−404	c_{18}	−1875690398545842	c_{34}	−289329469542493999278	c_{50}	70083483773215557
c_3	−2730	c_{19}	−2183462208838842	c_{35}	−354976961208562346202	c_{51}	−2115744906153941
c_4	−14615	c_{20}	20452918511924362	c_{36}	387441763374977804593	c_{52}	62240741999607227
c_5	469714	c_{21}	68856019739527374	c_{37}	−1122885795202543698132	c_{53}	51176341495631928
c_6	1419637	c_{22}	−93656886597222765	c_{38}	458480249804434527704	c_{54}	15374568403383506
c_7	−12006056	c_{23}	−791918247127969414	c_{39}	951306463215010994748	c_{55}	54232498348067132
c_8	−178792414	c_{24}	−425463457343288922	c_{40}	5043371415568642542524	c_{56}	−417454139397018
c_9	−153440	c_{25}	4650800507162790736	c_{41}	2489697533811537836608	c_{57}	20564608554658910
c_{10}	6366125185	c_{26}	8823106676760904034	c_{42}	1275266834962788273288	c_{58}	−167834169849393
c_{11}	28171466424	c_{27}	−11136855929081869602	c_{43}	678167240075824081928	c_{59}	2200590812774165
c_{12}	−107999328837	c_{28}	−47770253366115521178	c_{44}	411174638026342236400	c_{60}	−350223777255459
c_{13}	−1145768291192	c_{29}	−38695590226233318312	c_{45}	1588136191356704062624	c_{61}	−102217930564787
c_{14}	−1172534675912	c_{30}	127697364835635102126	c_{46}	−847669318309190494464	c_{62}	−62793771318666
c_{15}	22989931514252	c_{31}	201819845042020005732	c_{47}	−5534252585046702269824	c_{63}	107869671805419
c_{16}	86446086563248	c_{32}	120992409866075169155	c_{48}	2354183978283850391680		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 63, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $63 + 4$ terms removed returns order 63 as well, so $\deg q_{\min} = 63 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $63 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 63 that this sequence satisfies — and that family turns out to have exactly one member.

References

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